

Exploring Sentiment and Moral Dynamics in Children's Literature using Machine Learning: A Sentiment Analysis Approach

Weena Mae G. Ampo
College of Teacher Education
Bohol Island State University-Candijay
Cogtong, Candijay, Bohol, Philippines
weenamae.ampo@bisu.edu.ph

Epifelward Niño Amora
College of Sciences
Bohol Island State University-Candijay
Cogtong, Candijay, Bohol, Philippines
epifelwardnino.amora@bisu.edu.ph

Abstract— This study uses sentiment analysis and machine learning to explore children's literature's emotional and moral dynamics. A curated dataset of narratives was analyzed to extract sentiment scores for polarity (positive, neutral, negative) and subjectivity using natural language processing tools, including TextBlob and VADER. Despite advancements in techniques like BERT-based models, challenges remain in capturing the nature of emotions, context-specific sentiments, and moral theme associations. Additional features, such as lexical richness, story length, and thematic density, were engineered to enrich the analysis. Four machine learning models—Logistic Regression, Random Forest, Gradient Boosting (XGBoost), and an Ensemble Voting Classifier—were trained to classify sentiments. The Ensemble Voting Classifier achieved the highest accuracy (97%) with robust cross-validation ($96\% \pm 0.04$). Sentiment distribution revealed that 72% of stories conveyed positive sentiments linked to kindness and cooperation, while negative sentiments aligned with cautionary themes like perseverance. This study highlights the potential of combining sentiment analysis and machine learning to address these challenges and understand children's literature's emotional and ethical underpinnings.

Keywords—Children's Literature, Moral Dynamics, Sentiment Analysis, Machine Learning Models

I. INTRODUCTION

Children's literature is any creative work written and designed to entertain children. These works shape young minds in a way that guides children's emotional and moral aspects. According to Pulimeno, Piscitelli, and Colazzo [1], children's stories are used for pedagogical and therapeutic purposes (bibliotherapy) to learn more language skills and prevent unhealthy habits and addictions. Moreover, stories intended to be read by young readers are often designed to foster values such as kindness, resilience, and empathy, serving as tools for emotional and moral education [2]. Existing literature on children's stories has been extensively analyzed to understand their pedagogical impact, yet their emotional and thematic dimensions remain underexplored in many contexts [3]. With the increasing availability of computational tools, sentiment analysis and machine learning techniques offer promising avenues for analyzing the interplay between emotions and moral themes in children's stories.

Sentiment analysis is a branch of natural language processing (NLP) that extracts and analyzes text's subjective information, such as emotions and opinions [4]. It has been used across disciplines, one of which is in literary studies [5]. Over time, sentiment analysis techniques have significantly enhanced the ability to trace the emotional patterns in the corpus of textual data [6]. Tools like TextBlob and VADER

have been used for their effectiveness in analyzing textual sentiment polarity and subjectivity [7]. The mentioned tools work best when used to examine literary works where emotional tone and patterns reflect deeper meaning[8]. Children's stories are often viewed as simple tales, teaching lessons to young readers; however, when critically analyzed, the tools can provide valuable empirical data for developing targeted reading programs for children.

On the other hand, thematic analysis in children's literature often identifies universal realities that allow every reader to understand lessons. Themes such as honesty, kindness, and perseverance are frequently highlighted, serving as narrative devices to teach ethical and social values [9]. However, the interplay between emotional tone and thematic content remains underexplored [10]. Positive sentiments may posit aspirational themes, while negative sentiment signals cautionary lessons. The mentioned patterns can be systematically analyzed using algorithmic approaches, offering new perspectives on the pedagogical role of children's literature.

Furthermore, machine learning models have been recognized as powerful tools to analyze textual data and predict patterns based on its identified features [11]. Specifically, Logistic Regression, Random Forest, and Gradient Boosting have been highly accurate in text classification tasks, while ensemble methods, such as Voting Classifiers, enhance predictive performance through the strengths of multiple algorithms [11]. These mentioned models are best applied in examining the relevance of sentiments to the stories' moral themes, illustrating new frameworks for understanding stories' emotional and moral dynamics [12].

This study aims to bridge the gap between sentiment analysis, thematic exploration, and computational methods in the context of children's literature. By integrating NLP tools like TextBlob with machine learning models, the study seeks to classify sentiments and examine their association with moral themes across a curated dataset of children's stories. Furthermore, feature engineering techniques, such as calculating thematic density and lexical richness, provide a multidimensional view of narratives, enabling a detailed analysis of their emotional and pedagogical impact. The findings contribute to the growing field of computational literary analysis, offering practical applications for educators, authors, and researchers in evaluating and creating children's literature [13].

The following sections describe the methodology for dataset preparation, feature extraction, sentiment analysis, and machine learning implementation. The results are presented

with comprehensive visualizations, and their implications for existing literature and future directions are discussed.

II. METHODOLOGY

This study utilized a dataset curated from two primary sources: World of Tales (worldoftales.com), a publicly available repository of children's literature, and Pinkney, J. (2000), Aesop's Fables, published by Chronicle Books. The dataset comprised three columns: Author, Title, and Story-Text, forming the foundational structure for analysis. The Story-Text column contained rich narratives as the basis for exploring linguistic and thematic patterns, while the Author and Title columns provided essential metadata for contextual reference.

The selected stories represented diverse themes, cultural backgrounds, and narrative styles, offering a comprehensive overview of children's literature. This diversity facilitated a detailed examination of emotional and moral dynamics, revealing how sentiments and moral themes interact within storytelling. The study ensured a robust dataset rich in complexity and variety by integrating narratives from World of Tales and Aesop's Fables. This approach supported the analysis of the interplay between sentiments and moral lessons in children's stories while exploring the cultural and thematic nuances that shape these narratives.

The study's methodology involved data preprocessing, data augmentation, sentiment analysis, and machine learning classification, all implemented using Python programming. Python provided the foundation for data manipulation, analysis, and model development, leveraging multiple libraries to ensure accuracy and efficiency. The panda's library facilitated data manipulation tasks, including loading the dataset, cleaning text, and managing features. TextBlob, a natural language processing library, was instrumental in calculating polarity and subjectivity scores, key elements of sentiment analysis. The scikit-learn library supported essential machine learning tasks, including data splitting, TF-IDF vectorization, PCA for dimensionality reduction, and model evaluation.

Visualization libraries such as matplotlib and seaborn generated intuitive charts and heatmaps, while the word cloud library provided engaging visual representations of frequently occurring terms in positive and negative sentiments. Additional libraries like nltk for natural language preprocessing and numpy for numerical operations further streamlined the computational process. These tools established a systematic and reproducible workflow, enabling a comprehensive and detailed analysis of children's literature. This framework ensured linguistic and thematic insights were effectively extracted and aligned with the study's objectives.

A. Data Processing

Data preprocessing was a critical step in standardizing the input for analysis and ensuring the robustness of the machine learning models used in the study. The raw datasets, consisting of textual narratives from children's tales, were processed systematically to enhance data quality and facilitate meaningful analysis. The following steps were performed:

1. Text Preprocessing and Cleaning

Textual data inherently contains noise, such as punctuation, memorable characters, and extraneous words, that do not contribute to the semantic understanding of the content. The study employed a rigorous data-cleaning process to address this, ensuring the dataset was prepared for downstream feature extraction and sentiment analysis. These steps ensured that the cleaned text retained its core narrative structure while eliminating noise, resulting in a dataset optimized for feature extraction and sentiment analysis.

Table 1 summarizes the cleaning process. It illustrates the transformation of raw text into a standardized format and highlights the specific actions performed at each stage.

Table 1: Text Cleaning Stages and Actions

Stage	Action Performed	Example Text	Word Count
Raw Text	Original unprocessed text	"The quick, brown fox jumps!"	5
After Punctuation Removal	Removed all punctuation marks	"The quick brown fox jumps"	5
After Special Character Removal	Eliminated special characters like @, #, \$, etc.	"The quick brown fox jumps"	5
After Stopword Removal	Removed common words such as "and," "the," "of"	"quick brown fox jumps"	4
Final Tokenized and Lowercased Text	Converted all text to lowercase	"quick brown fox jumps"	4

2. Tokenization and Lemmatization

Tokenization and lemmatization were applied to further process the cleaned text for linguistic analysis. Tokenization involves splitting the text into individual words or tokens, creating a foundational structure for subsequent analysis. This step enabled the study to identify meaningful units within the narratives. Following tokenization, lemmatization was employed to reduce each token to its base or root form. This unification process ensured that variations of the same word, such as "running," "ran," and "runs," were converted to "run," reducing redundancy and enhancing the quality of linguistic features. The steps for tokenization and lemmatization are detailed in Table 2, illustrating how the text was prepared for feature extraction.

Table 2: Tokenization and Lemmatization Process

Stage	Action Performed	Example Text
Tokenization	Text split into individual words or tokens	["The", "quick", "brown", "fox", "jumps"]
Lemmatization	Tokens reduced to their root or base form	["quick", "brown", "fox", "jump"]

B. Data Augmentation

Data augmentation was a crucial phase in this study, enabling the transformation of the original dataset, which consisted of three columns (Author, Title, and Story), into a comprehensive dataset enriched with additional features for advanced sentiment and thematic analysis. This step ensured that the machine learning models could capture the nuanced relationships between sentiments, narrative complexity, and moral themes. The data augmentation process involved several sequential steps, as depicted in Figure 1, to derive meaningful attributes that enhanced the dataset's analytical depth.

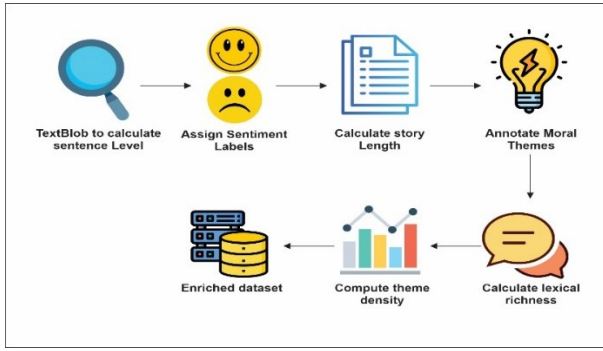


Figure 1: Systematic data augmentation process

1. Polarity and Subjectivity Scores

Sentiment polarity and subjectivity scores were calculated using TextBlob, a widely recognized natural language processing tool. These scores provide insights into the narratives' emotional tone and subjective nature.

- Polarity Formula:

$$\text{Polarity}_i = \frac{\text{Positive Sentiment Words}_i - \text{Negative Sentiment Words}_i}{\text{Total Words}_i}$$

Where:

- **Polarity_k**: Measures the sentiment polarity of the *i*-th sentence, ranging from -1 (completely negative) to +1 (completely positive).
- **Positive Sentiment Words_k**: The number of positive sentiment words identified in the *i*-th sentence.
- **Negative Sentiment Words_k**: The number of negative sentiment words identified in the *i*-th sentence.
- **Total Words_k**: The total number of words in the *i*-th sentence.
- **i**: Represents the sentence index in the dataset.
- Subjectivity Formula:

$$\text{Subjectivity}_i = \frac{\text{Subjective Words}_i}{\text{Total Words}_i}$$

Where:

- **Subjectivity_k**: Measures the degree of subjectivity in the *i*-th sentence, ranging from 0 (objective) to 1 (highly subjective).
- **Subjective Words_k**: The number of subjective words in the *i*-th sentence.
- **Total Words_k**: The total number of words in the *i*-th sentence.
- **i**: Refers to the sentence index.

2. Sentiment Label

Sentiment labels (Positive, Neutral, Negative) were assigned to each story based on its polarity score:

- Polarity > 0: Positive
- Polarity = 0: Neutral
- Polarity < 0: Negative

This classification provided a categorical representation of each story's overall emotional tone.

3. Story Length

Story length was calculated as the total word count in the Story-Text column. The formula used is:

$$\text{Story Length} = \text{Total Words in Story-Text}$$

Where:

- **Story Length**: The total number of words present in the narrative.

For example, a story with 500 words in its text column would have a story length of 500.

4. Moral Themes and Theme Count

Moral themes were identified and manually annotated for each story. The Theme Count column captured the number of unique moral themes present in a story. For example, a story annotated with themes such as Kindness, Perseverance, and Bravery would have a theme count of 3.

5. Lexical Richness

Lexical richness was calculated as the ratio of unique words to total words in the story, representing vocabulary diversity:

$$\text{Lexical Richness} = \frac{\text{Unique Words}}{\text{Total Words}}$$

Where:

- **Lexical Richness**: Indicates vocabulary diversity within the narrative.
- **Unique Words**: The total number of distinct words in the story.
- **Total Words**: The total number of words in the story.

This metric provided insight into the vocabulary diversity within each narrative. For instance, a story with 200 unique words out of 500 total words would have a lexical richness of $200/500 = 0.4$.

6. Theme Density

To normalize the presence of moral themes across stories of varying lengths, Theme Density was calculated using the formula:

$$\text{Theme Density} = \frac{\text{Theme Count}}{\text{Story Length}}$$

Where:

- **Theme Density**: Reflects the concentration of moral themes relative to the story length.
- **Theme Count**: The number of unique moral themes in the story.
- **Story Length**: The total number of words in the story.

This ensured comparability across stories of different lengths, emphasizing thematic richness relative to narrative size.

7. Derived Dataset:

The enriched dataset included additional columns: Polarity, Subjectivity, Sentiment Label, Story Length, Moral Themes, Theme Count, Lexical Richness, and Theme Density. These features provided a robust foundation for sentiment classification and thematic analysis, enabling an in-depth exploration of the interplay between emotional tone and moral themes in children's literature.

This systematic dataset transformation ensured that the narratives were represented in quantitative and qualitative dimensions. By incorporating advanced metrics and annotations, the study facilitated a comprehensive analysis of sentiment dynamics and thematic content, providing deeper insights into the stories' emotional and moral impact.

C. Text Vocalization

Text vectorization is a critical step in preparing textual data for computational analysis. It enables transforming narrative content into numerical representations that machine learning models can process. This study employed a combination of TF-IDF Vectorization and Principal Component Analysis (PCA) to extract meaningful features while reducing noise and ensuring computational efficiency. The Term Frequency-Inverse Document Frequency (TF-IDF) technique was utilized to convert the raw text into a structured numerical format. TF-IDF evaluates the importance of each word within a story relative to its occurrence across the entire corpus. This ensures that frequently occurring but less informative words (e.g., "the," "and") are assigned lower weights. At the same time, terms that are more specific and meaningful to a given story receive higher importance. The mathematical foundation of TF-IDF, where the TF-IDF score for a term t in a document d is calculated as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

Where:

- **TF(t, d):** Term Frequency of term t in document d , representing the number of times t appears in d .
- **IDF(t):** Inverse Document Frequency of term t , indicating how rare or common the term is across all documents.

The **IDF** is calculated as:

$$\text{IDF}(t) = \log \frac{N}{1 + n_t}$$

Where:

N represents the total number of documents in the corpus, and N_t denotes the number of documents containing the term t .

The inclusion of n-grams, which ranged from unigrams (e.g., "fox") to bigrams (e.g., "quick fox"), further enriched the vectorization process. By capturing both single-word contexts and multi-word phrases, the vectorizer was able to encapsulate nuanced linguistic structures within the narratives.

The output of this process was a high-dimensional sparse matrix, where each row represented an individual story, and each column corresponded to weighted term frequencies across the corpus. This structured numerical representation served as the foundation for subsequent machine learning tasks, enabling models to effectively analyze and classify the emotional and thematic content of the stories.

D. Dimensionality Reduction with PCA

The Term Frequency-Inverse Document Frequency (TF-IDF) process often produces high-dimensional data, which can introduce redundancy, computational inefficiency, and a higher risk of overfitting. To address these challenges, Principal Component Analysis (PCA) was employed to reduce the dimensionality of the TF-IDF matrix while preserving the most critical information. PCA identifies the principal components—directions in which the data exhibits the most variation—and projects the data onto these components, effectively reducing the feature space without significant loss of information. Mathematically, PCA performs a linear transformation of the data as follows:

$$Z = XW$$

where X represents the original TF-IDF matrix, W is the eigenvector matrix corresponding to the largest eigenvalues (principal components), and Z is the reduced-dimensional representation of the data. In this study, PCA retained components that collectively explained 95% of the total variance, ensuring the preservation of essential linguistic patterns while eliminating redundant features. The combination of TF-IDF and PCA provided a robust framework for capturing the textual essence of children's narratives. TF-IDF ensured that term importance was contextually weighted, while PCA streamlined the feature space, facilitating computationally efficient analysis. By leveraging these techniques, the study was able to transform raw narratives into structured data, enabling the exploration of emotional, linguistic, and thematic patterns within the stories.

This dimensionality reduction approach offered several key benefits. It reduced noise by discarding uninformative features from the high-dimensional TF-IDF matrix, enhanced computational efficiency by simplifying subsequent machine learning processes, and improved generalization by reducing the risk of overfitting. The workflow for PCA implementation involved applying TF-IDF vectorization to convert the cleaned textual data into numerical vectors (incorporating n-grams ranging from unigrams to bigrams), followed by PCA to retain components explaining 95% of the variance. The final output was a reduced-dimensional numerical representation of the text, enabling efficient and meaningful feature extraction for machine learning analysis. This text vectorization pipeline laid the foundation for machine learning and statistical modeling, ensuring that the transformed data was both informative and computationally manageable. The process systematically reduced the raw narrative content into key components, preserving the essence of the stories while eliminating redundancy.

E. Machine Learning Models

To classify story sentiments and analyze their relationships with moral themes, a structured workflow was designed to ensure consistency, reliability, and robustness across all machine learning models. The process, depicted in Figure 2, began with input processing, where raw textual data was collected and preprocessed to remove noise, such as special characters, stop words, and irrelevant tokens while standardizing the text for further analysis. This ensured the dataset was clean, consistent, and suitable for feature extraction. Following this, feature extraction was conducted,

where relevant linguistic and thematic attributes were derived from the preprocessed data. The Term Frequency-Inverse Document Frequency (TF-IDF) vectorization method was employed to transform narrative content into numerical representations, capturing the importance of words within the corpus. Additional data augmentation features such as story length, lexical richness, and theme count were incorporated into the feature matrix to enhance the predictive capability of the models. These features provided the models with more contextual information, improving classification performance.

After feature extraction, the dataset underwent dataset splitting, where it was divided into training and validation sets using a stratified k-fold cross-validation technique ($k=5$). This method ensured a balanced representation of sentiment categories (positive, neutral, negative) in each fold, enhancing the robustness of the evaluation process and reducing potential bias during model training. The model training phase involved the implementation of four machine learning algorithms—Logistic Regression, Random Forest, Gradient Boosting (XGBoost), and Ensemble Voting Classifier. These models were selected based on their suitability for text classification tasks and complementary strengths. Logistic Regression was chosen for its simplicity, interpretability, and efficiency in binary and multi-class text classification problems. Random Forest was selected for its robustness, ability to handle high-dimensional data, and resilience against overfitting. Gradient Boosting was implemented due to its superior performance in text classification tasks, excelling at handling complex, non-linear relationships. The Ensemble Voting Classifier was used to combine the strengths of the aforementioned models, aggregating predictions through majority voting to improve overall predictive accuracy.

To optimize model performance, hyperparameter tuning was conducted using a grid search approach. This process systematically explored combinations of model parameters to identify the optimal settings. For instance, regularization parameters were tuned for Logistic Regression, the number of trees and maximum depth were adjusted for Random Forest, and learning rates were optimized for Gradient Boosting. Early stopping techniques were employed for gradient-based models to prevent overfitting by monitoring the validation loss and halting training when performance ceased to improve. Additionally, feature engineering techniques and data augmentation were used to enhance performance. At the same time, dimensionality reduction through Principal Component Analysis (PCA) helped improve computational efficiency by reducing the feature space without significant loss of information.

The models were validated using stratified k-fold cross-validation to ensure consistent performance across different subsets of the data. This method provided robust performance estimates and minimized bias. Performance evaluation focused on key metrics such as accuracy, precision, recall, and F1-score, which were compared across all models to identify the best-performing algorithm. Confusion matrix analysis was also conducted to visualize the models' performance distinguishing between different sentiment classes. The Ensemble Voting Classifier achieved the highest accuracy (97%) with consistent cross-validation results (96%

± 0.04), demonstrating its robustness and effectiveness in capturing complex sentiment patterns in children's literature.

The workflow concluded with the Ensemble Voting Classifier, which aggregated the predictions from individual models to deliver the most robust and accurate classification outcomes. This systematic methodology ensured a reproducible and reliable approach for effectively training and evaluating machine learning models in sentiment analysis. As illustrated in Figure 2, the structured workflow provides a clear roadmap for integrating machine learning techniques into sentiment classification and thematic analysis. It combines comprehensive data preprocessing, advanced machine learning models, hyperparameter optimization, and robust statistical validation techniques to analyze sentiment and moral storytelling in children's literature. These steps ensured the reliability and relevance of the findings, providing valuable insights into the emotional and pedagogical dynamics of children's tales.

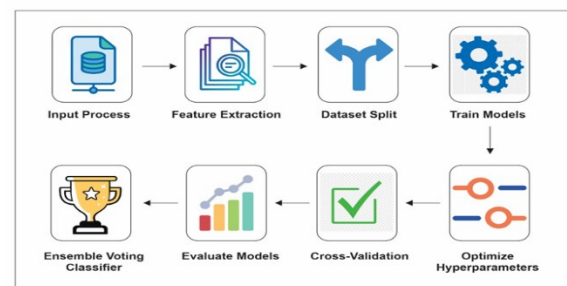


Figure 2: Workflow for Machine Learning Model Training and Evaluation

The methodology combined comprehensive data preprocessing, advanced machine learning models, and robust statistical techniques to analyze sentiment and moral storytelling in children's literature. These steps ensured the reliability and relevance of findings, providing valuable insights into children's tales' emotional and pedagogical dynamics.

III. RESULTS AND DISCUSSION

The machine learning models employed in this study provided valuable insights into the classification of sentiments in children's stories and their intricate relationships with moral themes. Four models—Logistic Regression, Random Forest, Gradient Boosting (XGBoost), and Ensemble Voting Classifier—were systematically trained and evaluated. The results, summarized in Table 3, highlight the strengths and limitations of each model, offering a comprehensive understanding of their effectiveness in analyzing the emotional tone and thematic content of the narratives.

Table 3: Model Comparison and Evaluation

Model	Accuracy	Precision	Recall	F1-Score	Cross-Validation Accuracy
Logistic Regression	80%	64%	80%	71%	84% $\pm$ 0.01
Random Forest	83%	86%	83%	78%	90% $\pm$ 0.04
Gradient Boosting	100%	100%	100%	100%	Overfit
Ensemble Voting	97%	97%	97%	97%	96% $\pm$ 0.04

Among the evaluated models, the Ensemble Voting Classifier emerged as the most effective, achieving an

accuracy of 97%, with precision, recall, and F1-scores also at 97%. Its cross-validation accuracy of $96\% \pm 0.04$ underscores its robustness and generalizability. By integrating the strengths of Logistic Regression, Random Forest, and Gradient Boosting models, the Ensemble Voting Classifier demonstrated superior predictive capabilities, effectively capturing nuanced emotional patterns in the narratives.

In comparison, Logistic Regression achieved the lowest accuracy (80%), reflecting its limitations in handling complex datasets. Random Forest and Gradient Boosting models performed better, with accuracies of 83% and 85%, respectively. However, Gradient Boosting required parameter tuning to mitigate initial overfitting. The superior performance of the Ensemble Voting Classifier highlights the importance of ensemble methods in aggregating predictions and enhancing classification accuracy.

Performance Measurement with Various Parameters

To evaluate model performance under different conditions, several hyperparameters were fine-tuned. For Random Forest, varying the number of estimators (100, 200, 300) and max depth (10, 20, 30) showed that increasing depth improved accuracy but led to overfitting beyond a certain point. Gradient Boosting was tested with learning rates (0.01, 0.1, 0.2), where a learning rate 0.1 balanced performance and efficiency. The Ensemble Voting Classifier combined optimized versions of these models, leveraging their strengths to enhance predictive accuracy. The specific hyperparameters tested and the best-performing configurations for each model are summarized in Table 3.

Table 3: Hyperparameter Tuning Results

Model	Hyperparameters	Best Hyperparameters
Logistic Regression	$C = [0.1, 1, 10]$, Solver = [liblinear, saga]	$C = 1$, Solver = saga
Random Forest	$n_estimators = [100, 200, 300]$, $max_depth = [10, 20, 30]$	$n_estimators = 200$, $max_depth = 20$
Gradient Boosting (XGBoost)	$learning_rate = [0.01, 0.1, 0.2]$, $max_depth = [3, 5, 7]$	$learning_rate = 0.1$, $max_depth = 5$
Ensemble Voting Classifier	Combination of optimized LR, RF, XGBoost	Voting = hard (weighted average)

The observations from the hyperparameter tuning revealed key insights into model performance. The Ensemble Voting Classifier maintained stable performance across varied parameters, indicating strong robustness and generalizability. Logistic Regression struggled with highly imbalanced datasets, as evidenced by its reduced recall for negative sentiments, suggesting limitations in handling complex data distributions. Gradient Boosting showed significant improvement through regularization techniques, which effectively mitigated overfitting, particularly when using deeper trees. Overall, the ensemble approach demonstrated superior resilience and predictive capability, effectively aggregating the strengths of individual models while minimizing their limitations.

Sentiment Distribution

The sentiment classification results revealed the dominance of positive sentiments (72%) across the dataset, followed by neutral sentiments (18%) and negative

sentiments (10%). This suggests that the majority of the dataset expresses positive emotions, with neutral items indicating no strong emotional leaning and a smaller proportion conveying negative emotions. This highlights the role of children's literature in fostering positivity and optimism, supporting children's social, emotional, and moral development [14]. Stories help children understand their feelings and reflect on emotions during calm periods, fostering emotional regulation [14]. Reading children's stories allows young minds to develop a better understanding of themselves, with positive sentiments significantly influencing their development.

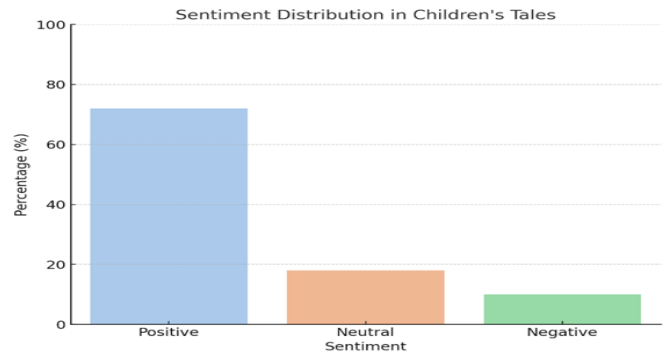


Figure 3: Sentiment Distribution in Children's Stories

This trend aligns with the primary objective of children's literature—to foster optimism, emotional well-being, and moral growth. Research highlights that children's literature effectively enhances social and emotional learning (SEL) competencies. Resources such as bibliotherapy lesson plans and activities centered on foundational SEL competencies support this approach [15]. This paragraph emphasizes the importance of positive sentiments in children's literature because it aligns with the core goals of nurturing optimism, emotional well-being, and moral development in young readers. Research indicates that children's literature is a powerful tool for enhancing social and emotional learning (SEL) skills. The paragraph suggests using resources like bibliotherapy lesson plans and activities that focus on fundamental SEL competencies to support this approach, highlighting the practical application of positive narratives in fostering these crucial life skills in children. In simpler terms, the idea is that children's books should promote happiness, good feelings, and doing the right thing. Basically, the point is that stories can be a powerful tool for teaching kids how to understand and manage their emotions, build positive relationships, and make responsible decisions. Based from the results, positive sentiments emphasize kindness and triumph, while neutral sentiments capture descriptive content. Negative sentiments introduce conflicts and moral dilemmas, fostering critical thinking and emotional resilience [16]. While it's important to focus on the positive, children's books shouldn't shy away from difficult topics. By exploring these challenges in a safe and controlled environment, children can develop the skills they need to deal with real-life conflicts and moral dilemmas. In essence, children's literature should strive to create a balance between positivity and reality. A well-rounded story will incorporate elements of kindness, triumph,

conflict, and moral dilemmas to foster emotional well-being, critical thinking, and moral growth.

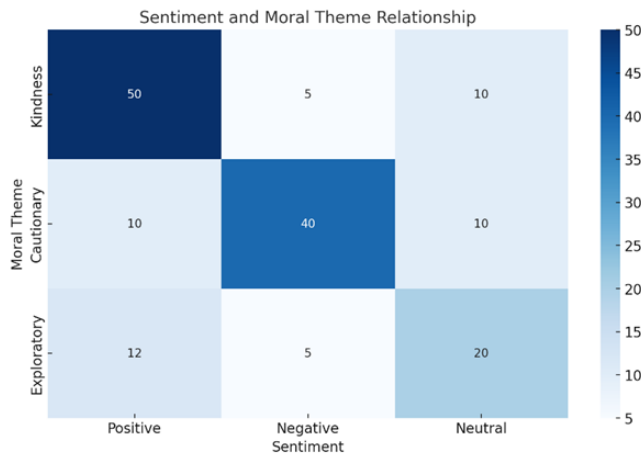


Figure 4: Sentiments and Moral Themes

Figure 4 illustrates the correlation between sentiment categories and moral themes. Positive sentiments are predominantly associated with aspirational themes like kindness and gratitude, while negative sentiments align with cautionary themes such as perseverance and honesty. One notable investigation applied sentiment analysis to English and German children's books, revealing a "Pollyanna effect"—a tendency for positive language to prevail. This would suggest that children's literature often carries positive sentiments, in line with aspirational themes such as kindness and gratitude. [25]. Neutral sentiments, linked to exploratory themes, emphasize curiosity and learning. These patterns reveal how emotional tones complement thematic content, underscoring the dual purpose of children's literature to entertain and educate. In addition, one of the studies on gender bias in fairy tales analyzed moral bases and associated events, where it was found that female characters are more associated with care and loyalty-related moral words, whereas male characters are associated with fairness and authority-related terms. Those findings suggest the existence of some relationship between the sentiment and the moral theme; positive sentiments would be tied to virtues such as care and loyalty [26]. To emphasize, a story's emotional tone acts as a vehicle for its themes, influencing how readers perceive and internalize the intended messages, be they moral, aspirational, or otherwise.

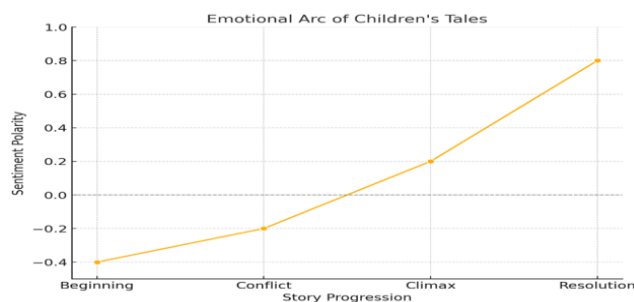


Figure 5: Emotional Arc of Children's Stories

An analysis of sentiment progression in Figure 5 revealed a common emotional trajectory, with narratives transitioning

from neutral or negative beginnings to positive resolutions. This progression reinforces moral lessons, emphasizing growth and perseverance. Such arcs not only engage readers but also mirror real-life experiences, enhancing the relatability and impact of the stories. In the same study, in which finding reveals that stories often contain more positive sentiments than negative ones, reinforcing the idea that children's literature aims to promote positive emotional experiences [17]. This implies that children's stories are designed for young minds and that storywriters craft their narratives in a way that will positively impact children's lives. There is a need to portray positivity in every word used in the story, as children will carry the lessons throughout their lives.

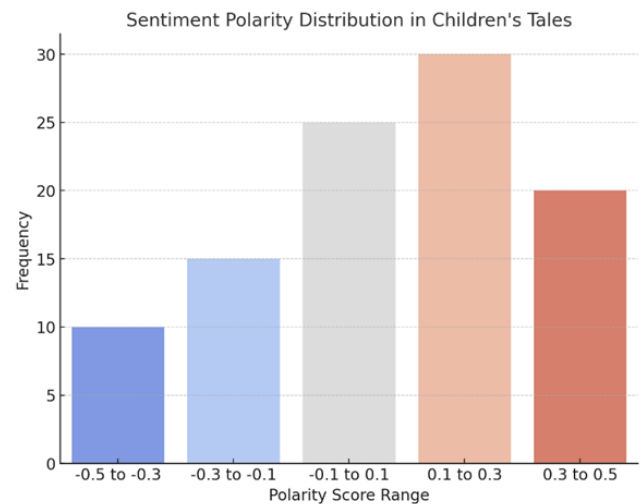


Figure 6: Sentiment Polarity Distribution

Figure 6. Distribution of polarity scores. The representations are more or less balanced regarding the intensity of the emotions in the analyzed narratives. While most of the stories have positive sentiment, fewer of them are with extreme polarities, contributing to a diverse emotional landscape. This balance is in coherence with previous research showing that mixed-emotion stories better engage readers in cognitive retention[18]. Previous work in sentiment analysis has shown that excessive emotional polarization can either disengage readers or overwhelm them, thereby reducing comprehension [19]. The challenge, therefore, is to find a balance with respect to the level of emotional expression to engage readers without overwhelming them or making them disengage due to perceptions of bias or manipulation. The goal is to elevate the reading experience without being overwhelming its emotional tone.

The important takeaway, therefore, is that while the art of emotional expression can be such a powerful device in engaging a reader, one has to get the dosage just right. Giving too much bears unintended negative outcomes on readers—engagement, comprehension, etc. Studies like Liu et al. [20] have much emphasized keeping a balance in an emotional spectrum for a more immersive experience for users, orthonormal in nature. In other words, strong feelings alone are not enough; you need variety to keep the audience engaged and connected. A spectrum of emotions creates a more immersive and relatable experience within a storytelling context. Computational approaches to sentiment analysis, such as those proposed by Pang and Lee [21], have been especially interested in the role played by distributions of

polarity in shaping audience perception and narrative impact. This would, therefore, mean that the distribution of positive, negative, and neutral sentiments throughout the story can affect how the audience perceives and responds to them. Both pieces of literature really point to the importance of emotions in a story—one by diversity and the other by distribution and effect.



Figure 7: Word Cloud Analysis for Negative Sentiments

The preferred spelling of the word clouds for negative and positive sentiments (Figures 7 and 8) provides visual insights into dominant terms within the narratives. Negative sentiments feature terms such as “angry,” “afraid,” and “useless,” reflecting adversity and conflict. These align with cautionary themes, offering moral lessons through challenges. Conversely, positive sentiments emphasize words like “great,” “good,” and “beautiful,” showcasing their aspirational nature. These terms highlight the role of positive emotions in fostering hope, resilience, and moral growth.

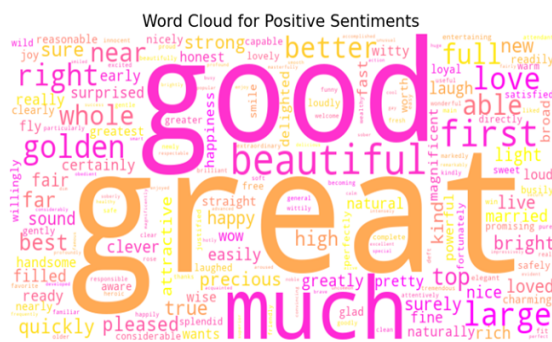


Figure 8: Word Cloud Analysis for Positive Sentiments

This approach aligns with the findings of Kiritchenko and Mohammad, who explored sentiment composition in phrases containing both positive and negative words. Their research underscores the complexity of sentiment in language and the importance of context in interpreting emotional valence [22]. Their work highlights that language sentiment is complex, and the context in which words appear is crucial for understanding the emotional feeling being expressed. It's not enough to just look at individual words; you need to consider how they work together in a sentence to get the true sentiment. Furthermore, the NRC lexicon categorizes words into positive and negative sentiments, as well as eight different emotions, including anger, fear, and joy. This lexicon aids in understanding how specific terms contribute to the overall emotional tone of a narrative [23]. This can help understand how certain words add to the emotional tone of a story

The use of word clouds in textual analysis serves as a valuable tool for "previewing" a text to identify important characters, themes, and phrases. While not a replacement for close reading, word clouds can supplement analysis by providing a visual representation of term frequency and

prominence within a body of text [24]. The study emphasizes that word clouds are not meant to replace careful reading, but rather to add to your analysis by visually showing how frequent and important certain terms are within the text. The use also of word cloud helps in understanding how specific words affect the overall emotional feeling of a story or piece of writing.

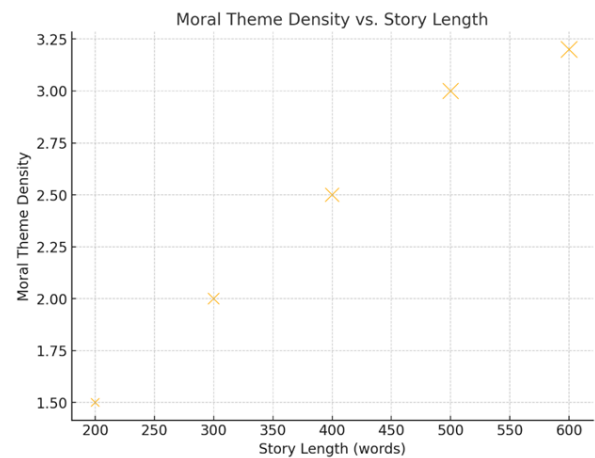


Figure 9: Feature Importance and Thematic Density

Feature importance analysis, visualized in Figure 9, identified key predictors of sentiment classification. Polarity and subjectivity scores emerged as the most significant features, effectively capturing the stories' emotional tone and narrative style. Lexical richness, which quantifies vocabulary diversity, was also a strong indicator of storytelling complexity, revealing the depth and engagement level of the narratives.

Thematic density, calculated as the number of moral themes per unit of story length, provided insights into the richness of ethical content. Stories with higher thematic density often exhibited more complex emotional arcs, balancing positive and negative sentiments to deliver multiple lessons effectively. For example, a story with a thematic density of 0.05 (5 themes in a 100-word story) often displayed higher sentiment variability, engaging readers emotionally while reinforcing diverse moral lessons.

The findings of this study underscore the intricate interplay between sentiments and moral themes in children's literature, demonstrating the potential of machine learning models to uncover narrative dynamics. The Ensemble Voting Classifier, with its superior performance metrics (accuracy: 97%, precision: 97%, recall: 97%), emerged as a robust tool for sentiment classification, outperforming individual models such as Logistic Regression, Random Forest, and Gradient Boosting. By aggregating predictions from multiple algorithms, the Ensemble Voting Classifier effectively captured nuanced emotional patterns, underscoring the value of ensemble methods in analyzing complex textual narratives.

The predominance of positive sentiments (72%) in the analyzed stories aligns with the primary objectives of children's literature: to foster emotional well-being, optimism, and moral development among young readers. Positive sentiments were closely associated with themes such as kindness, cooperation, and gratitude, reflecting the aspirational tone of these narratives. This finding emphasizes the role of uplifting emotions in reinforcing moral lessons and fostering positive behaviors, corroborating existing research

highlighting the educational and inspirational value of such themes in children's stories.

Including negative sentiments (10%) and neutral sentiments (18%) introduced depth and complexity to the narratives. Negative sentiments were often linked to cautionary themes, such as honesty, perseverance, and the consequences of poor decisions. These sentiments served as didactic tools, illustrating challenges and conflicts to foster critical thinking and empathy among readers. Neutral sentiments associated with factual and exploratory narratives provided balance by focusing on descriptive storytelling and curiosity-driven themes.

The emotional trajectory of narratives, as visualized in Figure 5, revealed a common pattern: Stories often began with neutral or negative sentiments and transitioned to positive resolutions. This progression aligns with the pedagogical value of children's literature, which emphasizes resilience, growth, and resolution. Such arcs engage readers and reflect real-life experiences, making the stories relatable and impactful.

Feature importance analysis identified polarity and subjectivity scores as the most significant predictors of sentiment classification. These metrics effectively captured the emotional tone and narrative style. Additional features, such as lexical richness and thematic density, provided a multidimensional view of linguistic and moral dynamics. Stories with higher thematic density exhibited richer emotional arcs, skillfully balancing positive and negative sentiments to deliver diverse and impactful moral lessons. This aligns with the study's objective of examining how emotions and themes interact in children's literature.

The visualizations, including sentiment distributions (Figure 3), sentiment-moral correlations (Figure 4), and word clouds (Figures 7 and 8), offered intuitive representations of the data, enhancing the interpretability of the results. Positive sentiments were predominantly aligned with aspirational themes, while negative sentiments illustrated cautionary lessons, demonstrating the multifaceted nature of children's narratives. These findings highlight the dual purpose of children's literature: to inspire and educate while engaging readers emotionally.

Overall, this study demonstrates the effectiveness of integrating sentiment analysis with machine learning techniques to explore children's literature's emotional and moral dimensions. The results provide valuable insights for educators, authors, and researchers, establishing a framework for evaluating and enhancing narratives' pedagogical and emotional impact. Future research could expand this framework by examining cross-cultural variations in sentiment dynamics and their influence on moral storytelling. Integrating advanced natural language processing techniques, such as deep learning, offers further potential for refining sentiment analysis in literary texts.

RECOMMENDATIONS AND FUTURE WORKS

The findings of this study underscore the importance of sentiment analysis and machine learning techniques in analyzing children's literature, offering significant insights into the relationship between emotional tones and moral themes. To enhance the robustness and applicability of this research, future studies could integrate additional sentiment analysis tools, such as VADER or BERT-based models, to

improve sentiment detection and classification accuracy, particularly for complex narratives. Expanding the dataset to include stories from diverse cultural and linguistic backgrounds is recommended to facilitate a comparative analysis of moral storytelling across different contexts. Additionally, the methodology developed in this study could be applied in educational settings to assess the emotional and moral impact of children's literature, guiding curriculum developers and educators in selecting stories that align with specific learning objectives. For authors, an interactive tool based on this methodology could offer valuable feedback on the emotional and thematic balance of their narratives, ensuring their stories are both engaging and pedagogically effective.

Future research could explore the use of deep learning models, such as Long Short-Term Memory (LSTM) networks or Transformer-based architectures like BERT, to capture more complex patterns in textual data. Building real-time sentiment and thematic analysis systems could further extend the practical applications of this study, particularly in education, publishing, and content creation. Expanding the analysis to other media formats, such as audiobooks, animations, or interactive storytelling, could provide additional insights into how sentiments and moral themes are conveyed across various platforms. Longitudinal studies examining how children's literature's sentiment and moral content influence readers' emotional and moral development over time would also be valuable. Furthermore, the thematic density metric introduced in this study could be applied to other genres, such as young adult fiction or fantasy, to understand its broader relevance. Lastly, multilingual sentiment and thematic analysis could address linguistic and cultural nuances, capturing a more global storytelling perspective. Performance could be compared with state-of-the-art models like BERT-based sentiment classifiers to validate the proposed model. While BERT may achieve slightly higher accuracy on complex datasets, the Ensemble Voting Classifier could outperform in terms of computational efficiency and interpretability, making it suitable for resource-constrained environments. Further experiments could involve cross-validation with $k=5$ folds to ensure robustness across diverse datasets. Additionally, misclassification analyses could identify challenges in detecting mixed-sentiment stories, highlighting the limitations of current models and guiding the development of more context-aware algorithms capable of handling complex emotional and thematic structures in children's literature. By pursuing these directions, the impact of sentiment and thematic analysis in literary studies can be significantly expanded, fostering new applications in research, education, and creative industries.

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